



Bioguided isolation, identification and bioactivity evaluation of anti-MRSA constituents from *Morus alba* Linn.

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ABSTRACT

Ethnopharmacology relevance: The root bark of *Morus alba* Linn. (*M. alba*), a traditional folk medicine, has been documented in the Chinese Pharmacopoeia, which has been widely used for asthma, fever, pneumonia, edema, vomit, colitis, bronchitis and keratitis diseases. Some of the diseases may be related to respiratory, digestive, urinary tract infections. Although Diels-Alder adducts (DAAs), flavonoids, 2-arylbenzofurans and stilbene compounds have been isolated from the root bark of *M. alba*, few compounds are reported for their antimicrobial efficacy *in vivo* and the mechanism.

Aim of the study: The aim of the study was to isolate and identify compounds of the root bark of *M. alba* in view of their anti-MRSA bioactivity, evaluate the anti-MRSA bioactivity of compounds and 60% ethanol elution (MA-6) *in vitro* and *in vivo*, and explore preliminary antibacterial mechanism in order to provide natural resources against MRSA infection.

Materials and methods: Systematic phytochemical investigations were carried out according to the thin layer chromatography (TLC) of the active fraction MA-6 to find more anti-MRSA ingredients. The compounds of the root bark of *M. alba* were separated by column chromatography and identified by LC-MS/MS and NMR spectroscopy. The anti-MRSA efficacy of the active ingredients were evaluated by broth microdilution method and a murine infection model. The mode of action of compounds was explored by time-kill curve and post-contact effect. The preliminary mechanism of compounds against MRSA was explored by drug efflux pumps and bacterial biofilms.

Results: Chemical isolation resulted in twenty-nine known compounds, most with one or more geranyl and prenyl units exhibited superior anti-MRSA bioactivity, with MIC values of 2–16 µg/mL. In addition, the mode of action indicated that compounds presented persistent antimicrobial effect, which also produced concentration-dependent and time-dependent killing activity or property. Preliminary mechanism showed that the compound kuwanon O (29) damaged the bacterial cell membranes, leading to the accumulation of antibiotics inside bacterial cells, moreover, MA-6 and kuwanon O (29) inhibited the efflux of drugs by combining with methicillin or ethidium bromide (EtBr), resulting in the MICs of EtBr and methicillin were obviously decreased three-fold. The anti-MRSA efficacy *in vivo* indicated that the active fraction MA-6 could reduce bacteria in spleen, liver, kidney and mortality of acutely infectious mice, which was better than the positive drug berberine chloride.

Conclusion: Experimental investigation showed that the MA-6 and compound 29 have promising bioactivity against MRSA *in vitro* and *in vivo*, which might be used as a potential source of new antibacterial medicine or a potential efflux pump inhibitor against MRSA infection.

1. Introduction

Infection in- and out-hospital caused by *Staphylococcus aureus* has

been a serious public health problem, which necessitates comprehensive measures in prevention and treatment. *S. aureus* is a clinical common pathogen that can cause illnesses ranging from mild skin and deep tissues to life-threatening infections such as pneumonia, bacteremia,

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Abbreviations

CCCP	carbonyl cyanide <i>m</i> -chlorophenylhydrazone
CFU	colony forming unit
CPZ	chlorpromazine
DAA	Diels-Alder adducts
DMSO	dimethyl sulfoxide
EtBr	ethidium bromide
EtOAc	ethyl acetate
EPI	efflux pump inhibitor
MA-6	60% ethanol elution of the root bark of <i>M. alba</i>
MBC	minimum bactericidal concentration
MIC	minimum inhibitory concentration
MRSA	methicillin-resistant <i>Staphylococcus aureus</i>
NMR	nuclear magnetic resonance
PBS	phosphate buffered solution
SEM	scanning electron microscope
TLC	thin layer chromatography

osteomyelitis and arthritis (Ross et al., 2008; Ku et al., 2011; Farooq et al., 2015; Que et al., 2018). With the extensive use of methicillin, *S. aureus* (MRSA) is resistant to methicillin by obtaining penicillin-binding protein 2a (PBP2a) (Jorgensen et al., 1991), which led to higher morbidity and mortality, as well as increased health care costs (Guillamet et al., 2018). Furthermore, the types of drug-resistant strains have gradually increased (Jonathan et al., 2012), such as drug-resistant *Mycobacterium tuberculosis*, drug-resistant *Pseudomonas aeruginosa* and drug-resistant *Escherichia coli* (Naoya et al., 2018). For a long time, vancomycin and teicoplanin have achieved effective anti-MRSA bioactivity by destroying the integrity of the cell wall, and they have been the last resort antibiotics against MRSA. However, large-scale use has led to the emergence of vancomycin-resistant MRSA (Wang et al., 2015).

The general mechanisms of bacterial resistance to antimicrobials include forming biofilms, activating the drug efflux pump-related proteins, altering the targeted drug sites and decreasing membrane permeability (Heim et al., 2020; Tintino et al., 2016). Among them, the formation of biofilms conferring higher tolerance to many antibiotics and to the host immune response (Cadet et al., 2013), which is one of the major obstacles in the fight against many bacterial infections. Moreover, long-term high dose combination antibiotic therapy has led to the reduction of biofilms rather than eradication (Wu et al., 2015), there are currently no effective clinical method to combat bacterial infections based on biofilms (Melander et al., 2020). Worst yet, the biofilm is formed through four steps of adhesion, microcolonies development, biofilm maturation, and dispersion (Archer et al., 2011; Beloin and Ghigo, 2008; Tsiry et al., 2015), which have an adhesin function, can colonize on the surface of the host by forming a rigid and stable structure, thereby escaping the phagocytosis of immune cells (Kristian et al., 2003). The biofilms of bacterial cells also exhibit altered gene expression of mediate antibiotic resistance, such as efflux pumps (Mantareva et al., 2011). Efflux pumps are the main mediators of resistance mechanisms to biofilm structure, it is a significant role in the cell-cell signaling involved in biofilm formation (Pannek et al., 2006; Hirakata et al., 2009). It is an effective way to overcome drug resistance by blocking the efflux of antibiotics, increasing the concentration of intracellular drug, thereby minimizing the ability of the biofilms formation of bacterial cells (Vasudevan et al., 2014; Sun et al., 2014). Therefore, our research was focused on developing potential efflux pump inhibitors (EPIs), to improve the susceptibility of MRSA toward antibiotics, reducing the MIC of the antibiotics (Chan et al., 2011; Mai et al., 2013). Therefore, it is necessary to search for a variety of sustainably effective chemical drugs from natural resources. Drug resistance is a strategy of survival competition among microorganisms, then antibiotics may exist with

drug resistance (Letten et al., 2021), however, the structure and anti-bacterial mechanism of plant natural products are quite different from antibiotics, a single compound could act on multiple targets, which may solve the problem of multi-drug resistance (Zuo et al., 2018). Therefore, we need to find the novel antibacterial agents from TCM based on their uses in folk.

Morus alba Linn is a plant of genus *Morus*, family distributed in Asia, Europe, Africa and South America (Yang et al., 2011). Phytochemical studies have shown that the root bark of *M. alba* contain many bioactive compounds, such as DAAs, flavonoids, 2-arylbenzofurans, stilbene compounds, and others (Carmin et al., 1983). The root bark of *M. alba*, a traditional chinese medicine (TCM), has been widely used for asthma, fever, pneumonia, edema, vomit, colitis, bronchitis and keratitis diseases, some of the diseases may be related to respiratory, digestive, urinary tract infections, and may be caused by MRSA infection (Zora et al., 2017). Previously pharmacological studies have shown that the root bark of *M. alba* has antibacterial bioactivity, but only the MICs and MBCs of a few compounds, crude extracts and fractions were reported (Ross et al., 2008; Ku et al., 2011; Farooq et al., 2015). The lack of systematic phytochemical investigation, antibacterial evaluation *in vitro* and *in vivo* against MRSA and preliminary mechanism investigation.

Through the study of the mode of action of active compounds against drug-resistant bacteria, it is expected to find a new anti-MRSA pathway on the basis of the study on bacterial biofilms and drug efflux pumps (Conlon et al., 2013). These studies could provide a theoretical and experimental basis for the treatment of chronic persistent intracellular drug-resistant bacterial infection and the destruction of bacterial biofilms by plant natural products. Therefore, our work was focused on screening anti-MRSA compounds, evaluating anti-MRSA efficacy of compounds *in vitro* and *in vivo*, and exploring preliminary antibacterial mechanisms based on bacterial biofilms and drug efflux pumps.

2. Materials and methods

2.1. General experimental procedures

Compounds were concentrated by vacuum rotary evaporators (OSB-2200 EYELA) and separated by column chromatography, including silica gel (200–300 mesh), lichroprep RP-C18 (40–63 µm), and sephadex HL-20 (40–70 µm). Compounds were purified by an Agilent 1260 Infinity II (USA) with diode array detector using a semi-preparative RP-C18 (250 mm × 10 mm, 5.0 µm). Thin layer chromatography (TLC) was performed on silica gel plates using silica gel GF 254 (HG/T2354-2010). ESI-MS spectra was recorded on an Agilent 1290/6545Q-TOF (Agilent, Santa, Clara, USA). NMR spectra was recorded on a Bruker AV-400 instrument and processed with MestReNova software. Ethyl acetate, petroleum ether, chloroform, methanol, acetone and acetonitrile were purchased from Tianjin Chemical Reagents Co.

2.2. Plant material

The dried root bark of *M. alba* (10 kg) was purchased from the herbal medicine market of Luosiwan, Kunming, Yunnan province, China, in May 2020. The herb was checked with <http://www.theplantlist.org>. and determined by Associate Professor Li-Xing Zhao of Yunnan University. A specimen (No. YD-377) has been stored in the Key Laboratory of Medicinal Chemistry for Natural Resource, Ministry of Education and Yunnan Province, School of Chemical Science and Technology, Yunnan University, Kunming, P.R. China.

2.3. Strains and reagents

Standard strain *S. aureus* (ATCC4330) and three clinical isolates of MRSA (MRSA-031, MRSA-011 and MRSA-003) were from the First People's Hospital of Zunyi City. Standard antibiotics ampicillin trihydrate, kanamycin sulfate, methicillin, and berberine chloride were

purchased from Macklin Reagents Co (Shanghai, China). Ethidium bromide (EtBr), standards efflux pump inhibitor chlorpromazine (CPZ), and carbonyl cyanide *m*-chlorophenylhydrazide (CCCP) were purchased from Macklin Reagents Co (Shanghai, China).

2.4. Animals

Eight-week-old female C57/BL6 mice (20 ± 2 g) were purchased from Kunming Medical University. Animals were reared in a light/dark cycle of 12 h, room temperature (about 25 °C) and a constant humidity (40%–60%). All animals were adapted to the laboratory environment for one week, which were allowed free access to water and food. All studies were conducted in laboratory with no specific pathogen grade. Animal studies were carried in accordance with international guidelines of animal experiments and internationally recognized ethical principles (the approval number: YNUCARE20210020).

2.5. Extract isolation

The dried root bark of *M. alba* (9 kg) was crushed and ultrasonically extracted in 95% ethanol solution at 55 °C for 3×0.5 h, with 80 L each time. The EtOH extracts were combined, filtered, and concentrated to obtain the crude extract (1.58 kg). In order to obtain effective active fractions quickly, a small amount of crude extracts were separated by macro-porous resin (D-101) column. The fractions with different polarity were obtained, and most of the remaining crude extracts were separated by referring to the TLC characteristics of the active fraction. Therefore, small doses of crude extract (1 g) was suspended in water and subjected to a D-101 column, eluted with a suitable gradient elution procedure of EtOH/H₂O (0:100, 20:80, 40:60, 60:40, 80:20, and 95:5 v/v, 1.0 L for each ratio) to generate six fractions (MA-0, MA-2, MA-4, MA-6, MA-8, MA-10). The crude extract and six dried fractions were dissolved in dimethyl sulfoxide (DMSO) for preliminary determination of anti-MRSA bioactivity by broth microdilution method, among them, the crude extract and MA-6 showed better anti-MRSA bioactivity, however, the other fractions showed inferior anti-MRSA bioactivity, therefore, for finding to more the anti-MRSA compounds. According to TLC and LC-MS of MA-6, the systematic phytochemical investigations of the crude extract (about 1.58 kg) of *M. alba* were carried out in this study, and the remaining crude extract was treated as Fig. S1. The compounds with similar TLC characteristics to MA-6 were obtained.

2.6. Identification of compounds from MA-6 by UHPLC-QTOF MS/MS

A previous study showed that MA-6 had better anti-MRSA bioactivity, so it is necessary to analyze the antibacterial substance of MA-6 through the UHPLC-QTOF MS/MS. The analysis was performed by an Agilent 1290 UHPLC instrument coupled and 6545 Q-TOF/MS system (Agilent Technologies). The samples were separated by using an Agilent SB-C18 column (2.1×100 mm, 1.8 μ m; Agilent Technologies) at a constant temperature (30 °C). The solvent system consists of water containing 0.1% (v/v) formic acid (A) and methanol (B) at a flow rate of 0.15 mL/min. The injection volume is 2 μ L. The gradient elution was 0–15 min, 5%–15% B; 15–25 min, 15%–30% B; 25–35 min, 30%–45% B; 35–60 min, 45%–70% B; 60–75 min, 70%–90% B; 75–80 min, 90%–98% B. The scanning range in the negative ion mode was *m/z* 100–1100. The ESI voltage in the negative ion mode was 3.0 kV. A nebulizing gas of 35 (psig) and a drying gas (8 L/min) were applied for ionization, using nitrogen in the negative ion mode. These conditions were used to identify compounds of MA-6. By comparing fragment ions with the literatures, and the retention time from MS and MS/MS data with the isolated compounds, compounds of MA-6 can be identified. By using MS, NMR and comparing with the data of literature to characterize the structure.

2.7. Evaluation of anti-MRSA bioactivity in vitro

2.7.1. Susceptibility testing

The antimicrobial activity of the isolated compounds was screened to determine the anti-MRSA compounds. Here, the minimal inhibitory concentrations (MICs) and minimal bactericidal concentrations (MBCs) of MA-6 and all compounds were measured by broth microdilution method (Qiu et al., 2018). The MBC was determined subsequent to MIC testing, was defined as the lowest extract concentration at which no bacterial growth was observed (Benincasa et al., 2015). The MBC was determined based on colony forming units (CFU) enumerated from tryptic soy broth (TSB) agar plates. Aliquots (10 μ L) from each well at the concentration of the MIC and the concentrations above were transferred into TSB agar plates and incubated at 37 °C for 12 h. The MBC was determined as the lowest concentration where no bacterial growth was observed (Danac and Mangalagiu, 2014). Positive controls (ampicillin trihydrate, kanamycin sulfate, methicillin, berberine chloride), a growth control, a sterility control, and the bacteria incubated in TSB containing 10% DMSO was used as blank controls. All experiments were repeated three times.

2.7.2. Time-kill analysis

The active compounds can be used for further evaluate whether the compound produced a concentration-dependent or time-dependent killing, it is conducive to the next research of mechanism and the evaluation of efficacy. As described previously (Marceau et al., 2013; Wang et al., 2020), in 24-well round bottom plates, time-kill curves were tested by using concentrations $1 \times \text{MIC}$, $2 \times \text{MIC}$ and $4 \times \text{MIC}$, respectively. First, added TSB media to 0.9 mL per well, then added MRSA-011 suspension (1×10^7 CFU/mL), and the final volume was 1.0 mL. The plates were incubated at 37 °C for 0, 1, 2, 3, 4, 6, 12, 16, 20, and 24 h. Aliquots (10 μ L) from each culture were serially diluted in sterile saline, and then moved to TSB agar plates, which were incubated at 37 °C for 24 h, then, the bacterial colonies were enumerated. All experiments were repeated three times.

2.7.3. Post-contact effect (PCE)

The time-kill curves determined whether the compound is a bactericidal or an antibacterial agent, thus, it is necessary to further confirm whether these agents have persistent inhibition on the growth rate of bacteria. Therefore, as described previously (Wang et al., 2015), in 24-well round bottom plates, PCE was tested by using concentrations $1 \times \text{MIC}$, $2 \times \text{MIC}$ and $4 \times \text{MIC}$, respectively. First, added TSB media to 0.9 mL per well, then added MRSA-011 suspension (1×10^7 CFU/mL), and the final volume was 1.0 mL. The plates were incubated at 37 °C for 3 h. The cultures were washed three times in sterile saline, and aliquots (10 μ L) from each culture were serially diluted in sterile saline, and then moved to TSB agar plates, which were incubated at 37 °C for 0, 1, 2, 3, 4, 6, 12, 24 h to determine CFU. All experiments were repeated three times.

2.7.4. Evaluation of efflux pump inhibition

The active efflux of MRSA was one of the main mechanisms of multi-drug resistance. Presently, it is an effective way to screen efflux pump inhibitors (EPIs) with no toxic side effects from plant natural products, thereby reversed bacterial resistance. Therefore, the evaluation of efflux pump inhibition was performed in this study. As previously described (Tintino et al., 2016), first, the MICs of the EPIs, ethidium bromide (EtBr) and compounds were measured by broth microdilution method (Qiu et al., 2018), second, the test of inhibition of the efflux pump were performed by using a sub-inhibitory concentration ($\text{MIC}/8$) of EPIs, kuwanon O (29) and MA-6. All samples were dissolved in 10% DMSO to obtain a stock solution of 6.40 mg/mL. In 96-well round bottom plates, a 100 μ L aliquot of bacterial cell suspension corresponding to 1×10^5 CFU/mL and together with EPI and compound were added to all wells except those for media controls. The final concentration of bacterial was 1×10^5 CFU/mL and the final concentration range of the extracts was

0.5–512 µg/mL, incubated at 37 °C for 24 h. The MICs of controls were evaluated using the antibiotics and EtBr alone.

2.7.5. Scanning electron microscope (SEM)

Bacterial biofilms are closely related to the formation of bacterial resistance, not only a cause of resistance, but also an important mean of anti-MRSA. SEM was used to observe the effect of active compounds on the structure of biofilms and qualitatively analyze the morphological changes. Therefore, compound kuwanon O (29) at the 1 × MIC, 2 × MIC concentrations was added to the culture of MRSA-011 suspension (100 µL, 1×10^7 CFU/mL), the final volume was 1.0 mL, and the plates were incubated at 37 °C for 3 h, after that, bacterial cells were washed three times in 0.1 M PBS and fixed with 2.5% glutaraldehyde (v/v) at 4 °C for 24 h. Next, the bacterial cells were dehydrated in a series of ethanol solutions (30, 50, 70, 90, and 100 v/v%) for 10 min each sequentially,

and then the ethanol was replaced by acetone. The bacterial cells were dried in liquid CO₂, coated with a layer of gold by cathodic spraying and observed by using the SEM, as previously described (Wang et al., 2019).

2.8. Murine model of intraperitoneal infection

In order to determine whether the efficacy of the root bark of *M. alba* in the deep MRSA -011 infection *in vivo* is consistent with the activity *in vitro*, a murine model of intraperitoneal infection was established. The experimental twenty-four mice (weigh 20 ± 2 g) were randomly divided into four groups with six mice in each group. In the first group, mice were orally fed 200 µL PBS, which was the negative control group (KB group). One day before bacterial infection, all mice were rendered temporarily neutropenic by immunosuppression with cyclophosphamide at 50 mg/kg to acquire the mice with low immunity (Conlon et al.,

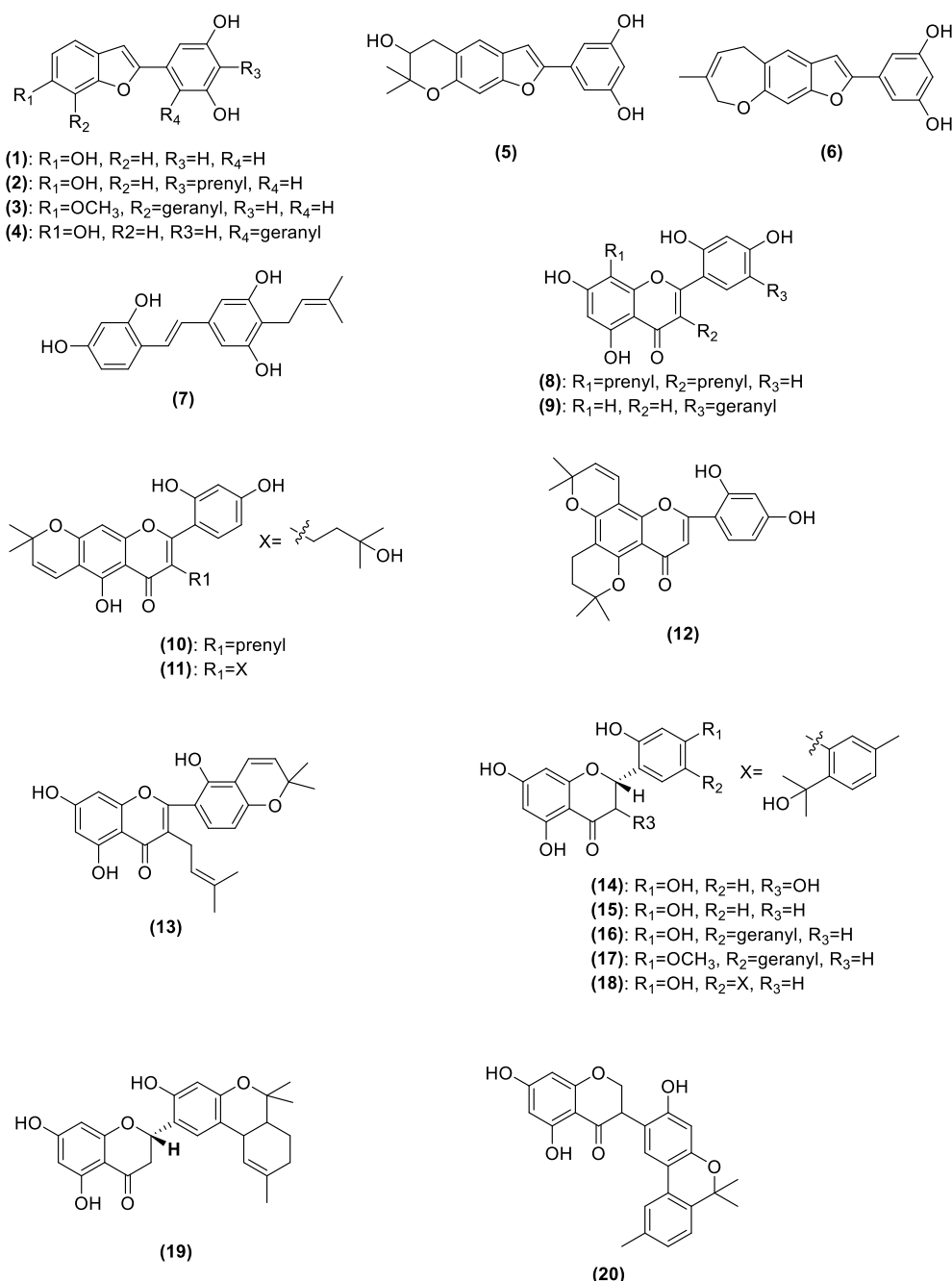


Fig. 1. Compounds isolated from the root bark of *M. alba*.

2013). The remaining three groups of mice were acutely infected with log-phase MRSA-011 by intraperitoneal injection with about 1×10^9 CFU/mouse. After 24 h, antibacterial treatment was initiated, mice were orally fed 40 mg/kg MA-6 (MA-6 group), 50 mg/kg berberine chloride (Berberine chloride group) and 200 μ L PBS (CK group) once a day. After five days, the spleens, livers, and kidneys of the mice were harvested, and all the organs were weighed and homogenized in sterile saline, and the final volume was 1.0 mL. Aliquots (20 μ L) from each sample were serially diluted in sterile saline, and then moved to TSB agar plates, which were incubated at 37 °C for 24 h to determine CFU, as previously described (Wang et al., 2016).

2.9. Statistical analysis

All operations are carried out in the biosafety cabinet (BSC). All data are presented using GraphPad Prism 5.0, and were analyzed by *t*-test, with *p* < 0.05 considered statistically significant. Results are expressed as means $\pm$ SD.

3. Results

3.1. Isolation of compounds from the root bark of *M. alba*

The twenty-nine known compounds were isolated and identified from the root bark of *M. alba* (Fig. 1), among them, six 2-arylbenzofurans were identified as moracin M (1) (Long et al., 2016), moracin C (2) (Yang et al., 2012), mulberrofuran B (3) (Luo et al., 2016), albafulan A (4) (Seong et al., 2009), moracin P (5) (He et al., 2014), moracin G (6) (Takasugi et al., 1979); 3',5',2,4-tetrahydroxy-4 (3-methyl-1-butenyl) stilbene (7) (Dai et al., 2006a,b); thirteen flavonoids as kuwanon C (8) (Dai et al., 2006a,b), 5'-geranyl-5, 7, 2', 4'-tetrahydroxyflavone (9) (Liu et al., 2018), morusin (10) (Ji et al., 2011), morusin L (11) (Zheng and Yu, 2012), austrone A (12) (Zheng et al., 2012), kuwanon B (13) (Dai et al., 2006a,b), dihydromorin (14) (Jiang et al., 2015), steppogenin (15) (Dong-Cheol et al., 2017), kuwanon E (16) (Yang et al., 2015), kuwanon U (17) (Yang et al., 2015), benzokuwanon E (18) (Zheng and Yu, 2012), dicyclokuwanon E (19) (Zheng and Yu, 2012), and tetrapterol A (20) (Iinuma et al., 1995); nine DAAs as mulberrofuran G (21) (Fukai et al., 1985), mulberrofuran K (22) (Fukai et al., 1985), albanol B (23) (Fukai et al., 1985), mulberrofuran Q (24) (Dai et al., 2006a,b), kuwanol A (25) (Taro et al., 1985), kuwanon G (26) (Abbas et al., 2014), kuwanon H (27) (Shen and Mao, 2001), kuwanon L (28) (Nomura et al., 1988), kuwanon O (29) (Hano et al., 1984) by comparison of the physical and chemical data to the reported literatures.

3.2. Phytochemical analysis

The analysis of UHPLC-QTOF MS/MS indicated that the main ingredients were DAAs in MA-6. The secondary mass spectrometry analysis of the compound gave a series of fragments, which can be used as a reference for identification of the same type of compound by LC-MS. In the negative ion mode, the peaks of MA-6 were identified as follows: sanggenon T (U1), kuwanon P (U2), kuwanon X (U3), sanggenon D (U4), sanggenon C (U5), sanggenon C, isomer (U6), mulberroside C (U7), sanggenol A (U8), kuwanon H (27), kuwanon L (28), sanggenon G (U9), kuwanon K (U10), kuwanon O (29), kuwanon G (26), mulberrofuran Q (24), mulberrofuran C (U11), mulberrofuran J (U12), mulberrofuran G (21), mulberrofuran I (U13), kuwanon C (8), morusin (10), sanggenol L (U14), mulberrofuran K (22), mulberrofuran B (3), among them, compound 3, 8, 10, 21, 22, 24, 26, 27, 28 and 29 have been isolated (Fig. 2, Table S1).

The MA-6 of root bark of *M. alba* was identified twenty-four compounds, the main ingredients in MA-6 were DAAs and the identified DAAs can be formally divided into the four types, such as 2-arylbenzofurans, prenylflavanones, stilbenes, chalcones. DAAs showed the common characteristic fragment losses of one or more resorcinol units

(C₆H₆O₂). For example, the peak U11 *m/z* 579.1665 and the peak U12 *m/z* 579.1660 generated main fragment ions at *m/z* 469.1247 [M-H-C₆H₆O₂]⁻, 359.1274 [M-H-2 $\times$ C₆H₆O₂]⁻ and 427.1037 [M-H-2 $\times$ C₆H₆O₂-C₃H₆]⁻, the peak these ions were characteristic to chalcone-prenylated 2-arylbenzofuran adducts, therefore, they were compared with characteristic fragment ions, and tentatively annotated as mulberrofuran C (U11), mulberrofuran J (U12) (Aelenei et al., 2019). The peak 26 *m/z* 691.2187, the peak 27 *m/z* 759.2818, the peak 28 *m/z* 625.1711, the peak 29 *m/z* 693.2349, and the peak U10 *m/z* 691.2140, which displayed predominant fragment ions at 581.1573 [M-H-C₆H₆O₂]⁻, 471.1073 [M-H-2 $\times$ C₆H₆O₂]⁻, 469.1358 [M-H-2 $\times$ C₆H₆O₂-H₂]⁻, 419.1382 [M-H-C₁₅H₁₂O₅]⁻, 353.1064 [M-H-C₅H₆]⁻ and 297.1027 [M-H-C₄H₈]⁻, all these ions were main to chalcone-prenylflavone adducts, which might be generated common sequential cleavages of resorcinol units, chalcone resorcinol units and prenyl units. In view of same characteristic fragment patterns as reported previously in the literature (Ristivojevi et al., 2019), therefore, they were compared with standards of retention times and characteristic fragment ions, and tentatively annotated as kuwanon G (26), kuwanon H (27), kuwanon L (28), kuwanon O (29) and kuwanon K (U10).

Based on analysis of fragment, under collision-induced dissociation, the fragmentation behavior of chalcone-prenylated 2-arylbenzofuran, chalcone-prenylflavone/prenylflavanone and chalcone-prenylated chalcone adducts producing the high-abundance fragment ions: [M-H-H₂]⁻, [M-H-H₂O]⁻, [M-H-C₆H₆O₂]⁻, [M-H-C₆H₆O₂-H₂]⁻, [M-H-C₆H₆O₂-CO₂]⁻, [M-H-C₆H₆O₂-C₅H₁₀]⁻, [M-H-2 $\times$ C₆H₆O₂]⁻, [M-H-2 $\times$ C₆H₆O₂-C₃H₆]⁻, [M-H-C₅H₆]⁻ and [M-H-C₄H₈]⁻ (Aelenei et al., 2019). The analysis of fragment ions of compounds of MA-6 combined with the anti-MRSA bioactivity indicated that the predominant characteristic fragment ions, such as chalcone resorcinol units and prenyl units, playing an important role in anti-MRSA bioactivity. Therefore, superior anti-MRSA bioactivity of MA-6 was attributed to MA-6 contains many compounds with one or more chalcone resorcinol units, prenyl units and geranyl units.

3.3. Anti-MRSA bioactivity in vitro

3.3.1. Determination of MICs and MBCs

The results of MICs and MBCs indicated that the root bark of *M. alba* contain anti-MRSA compounds. Analysis of anti-MRSA bioactivity of all compounds were carried out at different concentrations (1–128 μ g/mL). The MIC of MA-6 was 32–64 μ g/mL. The results showed that all MRSA strains were resistant to methicillin (MIC $\geq$ 128 μ g/mL). Strongest anti-MRSA bioactivity as shown by moracin C (2), mulberrofuran B (3), albafulan A (4), kuwanon C (8), 5'-geranyl-5, 7, 2', 4'-tetrahydroxyflavone (9), morusin (10), kuwanon B (13), kuwanon E (16), and kuwanon H (27), which were showed better anti-MRSA bioactivity than the positive drugs kanamycin and ampicillin, with MICs values of 2–4 μ g/mL, and MBCs values of 2–8 μ g/mL. Strong anti-MRSA bioactivity as shown by 3',5',2,4-tetrahydroxy-4(3-methyl-1-butenyl) stilbene (7), kuwanon U (17), mulberrofuran G (21), mulberrofuran K (22), albanol B (23), kuwanon G (26), and kuwanon O (29), which were showed better anti-MRSA bioactivity than the positive drugs ampicillin and berberine, with MICs values of 4–16 μ g/mL, and MBCs values of 8–32 μ g/mL. Weak anti-MRSA bioactivity as shown by moracin G (6) and mulberrofuran Q (24), however, which were showed better anti-MRSA bioactivity than the positive drugs berberine, with MICs values of 8–16 μ g/mL, and MBCs values of 16–32 μ g/mL. Among them, the anti-MRSA bioactivity of albafulan A (4), moracin G (6), 3',5',2,4-tetrahydroxy-4(3-methyl-1-butenyl) stilbene (7), kuwanon B (13), mulberrofuran Q (24) and kuwanon O (29) were discovered for the first time (Table 1).

Geranylated 2-arylbenzofuran mulberrofuran B (3), albafulan A (4) and prenylated 2-arylbenzofuran moracin C (2) showed the greatest anti-MRSA bioactivity, whereas the non-prenylated compound moracin M (1) and moracin P (5) were not active. When the substitution group of

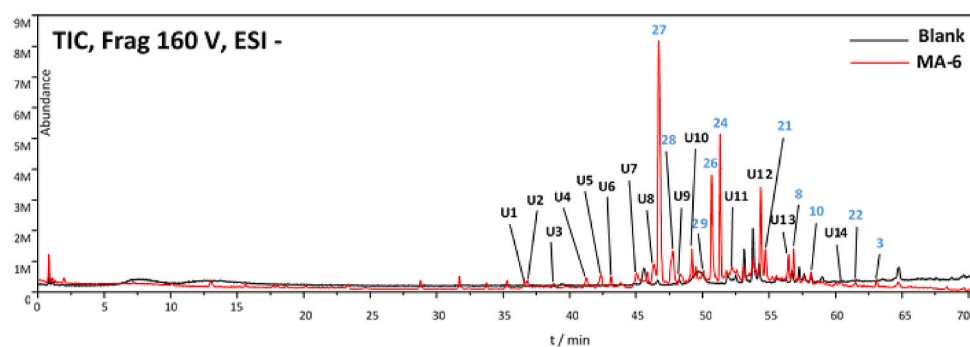
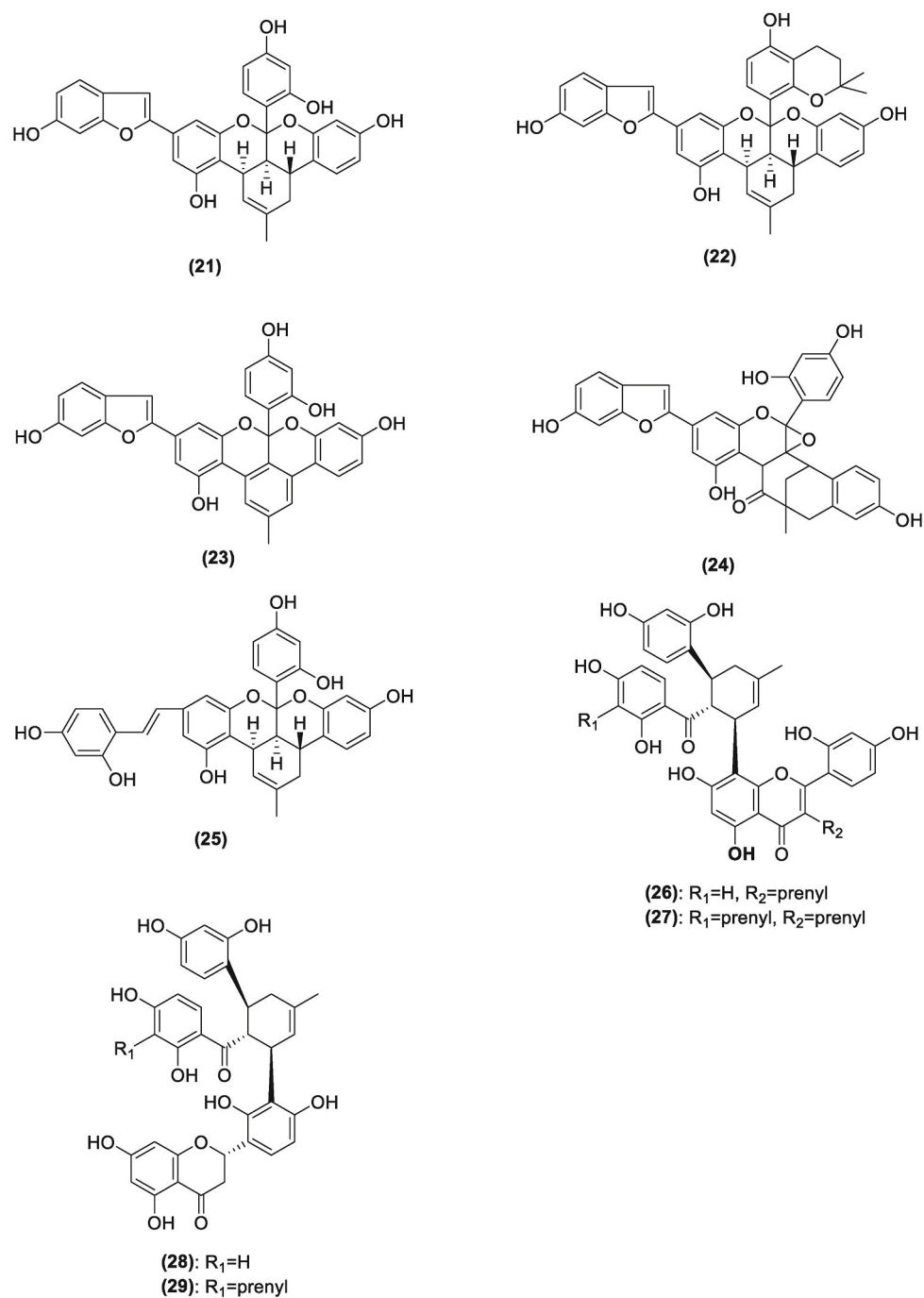


Fig. 2. In the negative ion mode, MS total ion current (TIC) chromatogram of MA-6. The black numbered peaks (U1–U14) are predicted compounds, and the blue numbered peaks are isolated compounds. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1The results of anti-MRSA bioactivity of MICs/MBCs ($\mu\text{g/mL}$).

Samples	MRSA-031		MRSA-011		MRSA-003		<i>S. aureus</i> ATCC4330	
	MIC	MBC	MIC	MBC	MIC	MBC	MIC	MBC
MA-6	32	64	32	64	64	128	32	32
2	8	16	8	8	8	16	8	8
3	4	4	2	4	4	8	2	4
4	4	4	2	4	4	8	2	4
6	8	16	16	32	16	32	8	16
7	8	16	4	8	8	16	4	8
8	4	8	2	4	4	4	2	4
9	2	4	4	4	4	8	2	4
10	8	8	2	4	4	8	4	4
13	4	8	4	4	4	8	4	4
16	4	8	4	4	4	8	4	8
17	4	8	8	8	4	8	4	8
21	8	16	8	16	16	32	8	16
22	8	16	4	8	8	16	4	8
23	8	16	8	16	16	32	8	8
24	16	32	8	16	16	32	8	16
26	4	4	4	4	4	8	2	4
27	2	4	2	2	2	4	2	2
29	8	16	16	16	8	16	4	8
Kanamycin	4	8	4	4	4	8	2	4
Ampicillin	16	32	8	16	16	32	4	16
Methicillin	256	512	128	256	256	512	32	64
Berberine	128	256	128	256	256	512	64	128

2-arylbenzofurans was 2'-geranyl, 7-geranyl, and 4'-prenyl, marked enhancement of effective antibacterial bioactivity was observed. Geranylated flavone 5'-geranyl-5, 7, 2', 4'-tetrahydroxyflavone (9), kuwanon E (16) and kuwanon U (17) showed the greater activity than prenylated flavone kuwanon C (8), morusin (10) and kuwanon B (13), however, the non-prenylated compound morusin L (11), austrone A (12) and steppogenin (15) was not active. DAA mulberrofuran G (21), mulberrofuran K (22), mulberrofuran Q (24) and albanol B (23) showed strong anti-MRSA bioactivity, moreover, DAA kuwanon H (27), kuwanon G (26) and kuwanon O (29) with a prenyl moiety showed strongest anti-MRSA bioactivity. When the substitution group of flavones was 5'-geranyl, and 3-prenyl showed superior antibacterial bioactivity. Previous studies have also confirmed the anti-MRSA potential of the root bark of *M. alba* constituents, such as compound moracin C, mulberrofuran B, mulberrofuran G, albanol B (Fukai et al., 2005), and kuwanon C, morusin, kuwanon E, kuwanon U, kuwanon H (Ulenová et al., 2019), which had obvious antibacterial or bactericidal effects on the clinically isolated strains MRSA, with MICs values of 1–25 $\mu\text{g/mL}$.

Our study identified that plant natural products with antibacterial bioactivity were greater than the drugs that are routinely used in clinical practice. Furthermore, these results indicated that prenylation or geranylation plays a crucial role in the anti-MRSA bioactivity of 2-arylbenzofurans, flavonoids and DAAs. Compounds with a geranyl moiety were greater bioactivity than the prenylated compounds, while the non-prenylated compounds were not bioactivity.

3.3.2. Time-kill assay

We need to have a deeper understanding of the against microbe mode of substance, carried out the test of MA-6, albafrican A (4), moracin G (6), kuwanon B (13), mulberrofuran Q (24) and kuwanon O (29) against MRSA-011 for the first time. This experiment is used to judge whether the compounds produce concentration-dependent or time-dependent killing and further judge whether the compound is a bactericidal or an antibacterial agent (Castro et al., 2013; Wistrand-Yuen et al., 2020). The time-kill curves were represented (Fig. 3), when the concentration of compound albafrican A (4), kuwanon B (13) and kuwanon O (29) was increased to $4 \times \text{MIC}$, the number of bacteria dropped to uncountable concentration, the bactericidal killing observed in about 3 h, and when the concentration of compounds was increased to

$2 \times \text{MIC}$, the bactericidal killing also observed in about 4 h. Therefore, these compounds had rapid bactericidal effect, produced concentration-dependent killing, the bactericidal effect was in a dose-response relationship. Compound moracin G (6), mulberrofuran Q (24) and component MA-6 at $4 \times \text{MIC}$, with bactericidal killing observed at 8 h. Additionally, cells treated with compound moracin G (6), mulberrofuran Q (24) and component MA-6 at the concentration of $1 \times \text{MIC}$ and $2 \times \text{MIC}$ did not exhibit killing within 12 h. These results indicated that compounds 4, 13 and 29 were bactericidal agents and they produced concentration-dependent killing; component MA-6, compounds 6 and 24 were antibacterial agents and they produced time-dependent killing.

3.3.3. Evaluation of post-contact effect (PCE)

We need to further assess whether the antimicrobial efficacy of the compound is persistent, we tested its post-contact effect (PCE) against MRSA-011. PCE was derived from the post-antibiotic effect, which was originally used to detect the persistence of suppression of microbial growth after exposure to drugs (Wistrand-Yuen et al., 2020; Odenholt et al., 1989). According to LC-MS analysis and time-kill assay results, MA-6, compound (4), compound (13), and compound (29) were evaluated on post-contact effect (Fig. 4).

The component MA-6, the compound albafrican A (4), kuwanon B (13), and kuwanon O (29) at the concentration of $1 \times \text{MIC}$, $2 \times \text{MIC}$ and $4 \times \text{MIC}$ were exposed to MRSA-011, the bacteria survival counts were decreased after 3 h. When the concentration of compound albafrican A (4) and kuwanon O (29) were increased to $1 \times \text{MIC}$, PCE values were both about 2 h, and when the concentration of compounds was increased to $2 \times \text{MIC}$ and $4 \times \text{MIC}$, PCE values were both 3 h and 4 h, respectively. Similarly, when the concentration of MA-6 and kuwanon B (13) were increased to $1 \times \text{MIC}$ and $2 \times \text{MIC}$, PCE values were both about 2 h, and about 3 h at $4 \times \text{MIC}$. All tests showed that the growth of bacteria was sensitive slowing after exposure to agents, and the growth of bacteria was accelerated after 5 h, but the CFU of the pretreated groups were lower than the untreated group within 24 h. These results indicated that constituents of the root bark of *M. alba* have effective anti-MRSA bioactivity and the data of PCE confirmed that MA-6, albafrican A (4), kuwanon B (13), kuwanon O (29) had persistent inhibition on the growth rate of bacteria.

3.3.4. Evaluation of efflux pump inhibition by MIC reduction

Bacteria is resistant to inhibitory substances such as toxic compounds and antibiotics, which is a mechanism of active efflux, known simply as drug efflux pumps (Vasudevan et al., 2014). Common multi-drug resistance bacteria overexpress efflux pumps, resulting in the efflux of many antibiotics and reduction the concentration of intracellular drug, thus, our research was focused on developing efflux pump inhibitors (EPIs) (Sun et al., 2014). Antibiotics with targeted EPIs can obstacle the efflux of antibiotics, decrease the invasiveness of bacteria as well as reduce the MIC of the antibiotics, and interrupt the formation of biofilms, thereby overcoming the problem of multi-drug resistance (Gupta et al., 2017).

Here, to assess whether compounds of the root bark of *M. alba* are potential EPIs, carried out a comparative study that compared between the antibacterial effects of standard EPIs and compounds, thereby evaluating the ability of both in reducing the MICs of ethidium bromide (EtBr) and methicillin. EtBr is a substrate for many multi-drug resistance pumps. The results of experiment showed as Fig. 5 and Table 2, when the MA-6 or kuwanon O (29) was combined with methicillin or EtBr against MRSA-011, synergistic effects were observed, and MICs of EtBr and methicillin were obviously decreased in three-fold. However, this strain lacks the EPIs, therefore, the enhancement of the methicillin anti-MRSA bioactivity can be caused by MA-6 and kuwanon O (29) via inhibition of the efflux of antimicrobials and the synergic effect can be attributed to this inhibition of mechanism. Similarly, the synergic effect was also observed when EPI was associated with the methicillin or EtBr against MRSA-011, indicating MA-6 and kuwanon O (29) against the efflux

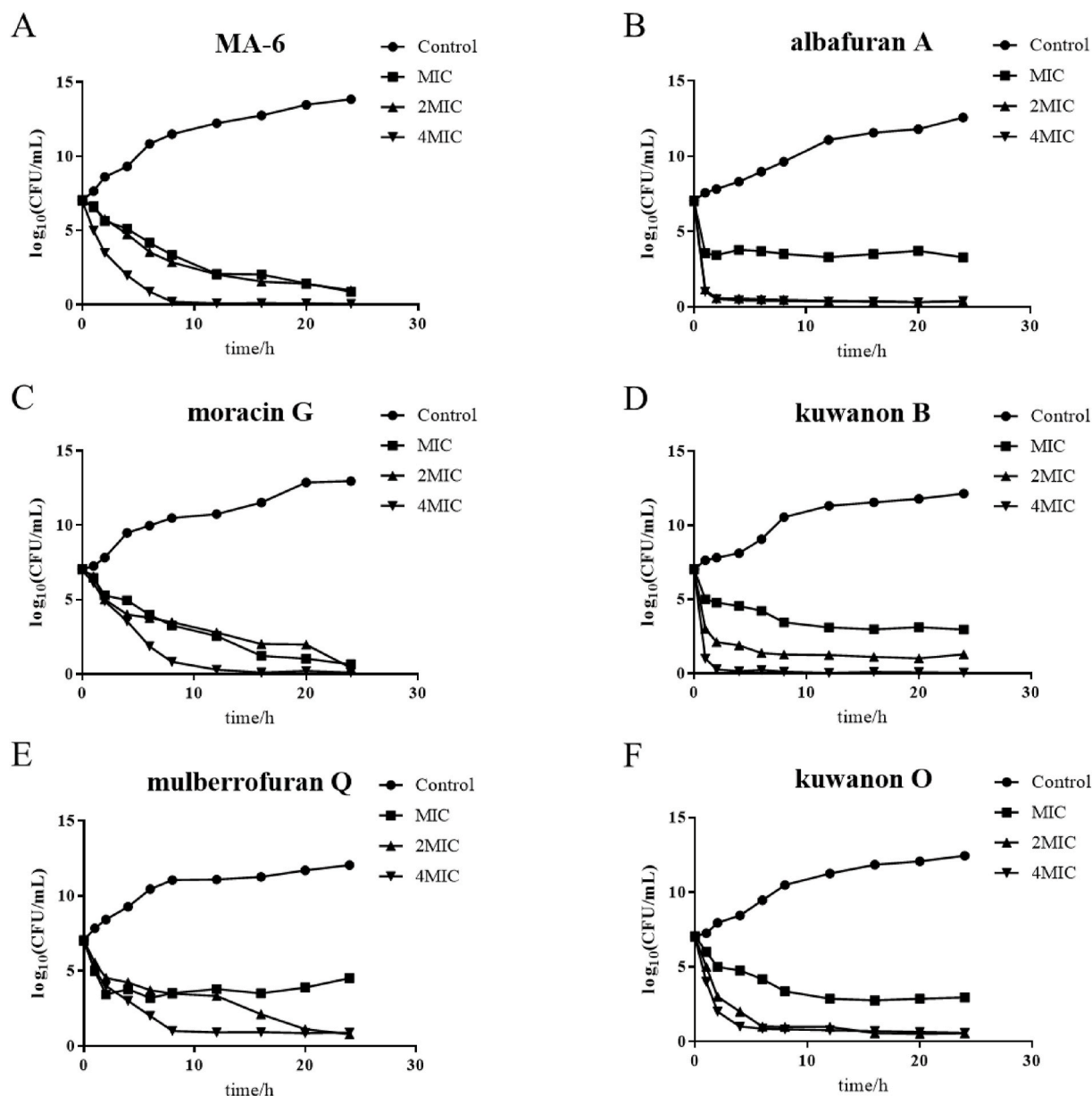


Fig. 3. Time-kill assays. Time-kill curves performed in TSB at 1 $\times$ MIC, 2 $\times$ MIC, 4 $\times$ MIC levels of component MA-6 (A), compound albafrican A (B), moracin G (C), kuwanon B (D), mulberrofuran Q (E) and kuwanon O (F) against the MRSA-011.

pumps of MRSA-011 have inhibitory effect, and was consistent with the effect of other pump inhibitors. The discovery of substances with the capacity to inhibit this mechanism is meaningful, therefore, new efflux pump inhibitors are promising compounds. Our results indicated that MA-6 and kuwanon O (29) had the same properties of the EPIs, thus possibly develop into a potential efflux pump inhibitor or an adjuvant in antibiotic therapy against multi-drug resistant bacteria.

3.3.5. Scanning electron microscope (SEM)

The formation of bacterial biofilm is an important reason for bacterial resistance to antimicrobials. The formation of biofilms is a major factor in the devastating complication of many bacterial infections, such as prosthetic joint infection (PJI) (Heim et al., 2020; Percival et al., 2015), it is difficult to treat because of them exhibited distinct metabolic that can be conducive to their chronicity and tolerance to antibiotics (Balaban et al., 2019; Cameron et al., 2018). Therefore, it is necessary to explore active natural product small molecules that interfered with the formation of bacteria biofilms, to reverse antibiotic tolerance endowed

by the biofilms.

According to results of efflux pump inhibition, we need to evaluate the morphological changes of bacterial cells following exposure to the compound kuwanon O (29) at the 1 $\times$ MIC (16 $\mu\text{g/mL}$), 2 $\times$ MIC (32 $\mu\text{g/mL}$) concentrations for 3 h by the SEM (Fig. 6). Untreated MRSA-011 cells (Fig. A1, B1) apparently exhibited regular, smooth and intact surface, however, after exposure to kuwanon O (29) at the concentration of 1 $\times$ MIC for 3 h (Fig. A2), the cell permeability was increased leading to dramatical morphological alternations of shape and size of cells, most of cell shapes changed from globular to irregular, deformed, wrinkled and sunken. Similarly, after treatment with kuwanon O (29) at the concentration of 2 $\times$ MIC for 3 h (Fig. A3), the morphology of bacteria obviously changed and deformed, even resulting in the leakage of contents, the destruction of bacterial biofilm and death of almost all cells. After exposure to the positive drug methicillin at the 1 $\times$ MIC (256 $\mu\text{g/mL}$), 2 $\times$ MIC (512 $\mu\text{g/mL}$) concentrations for 3 h (Fig. B2, B3), the membrane morphology of cells became pitted, irregular and shrunk, however, the effect was worse than kuwanon O (29). Bacterial biofilms are closely

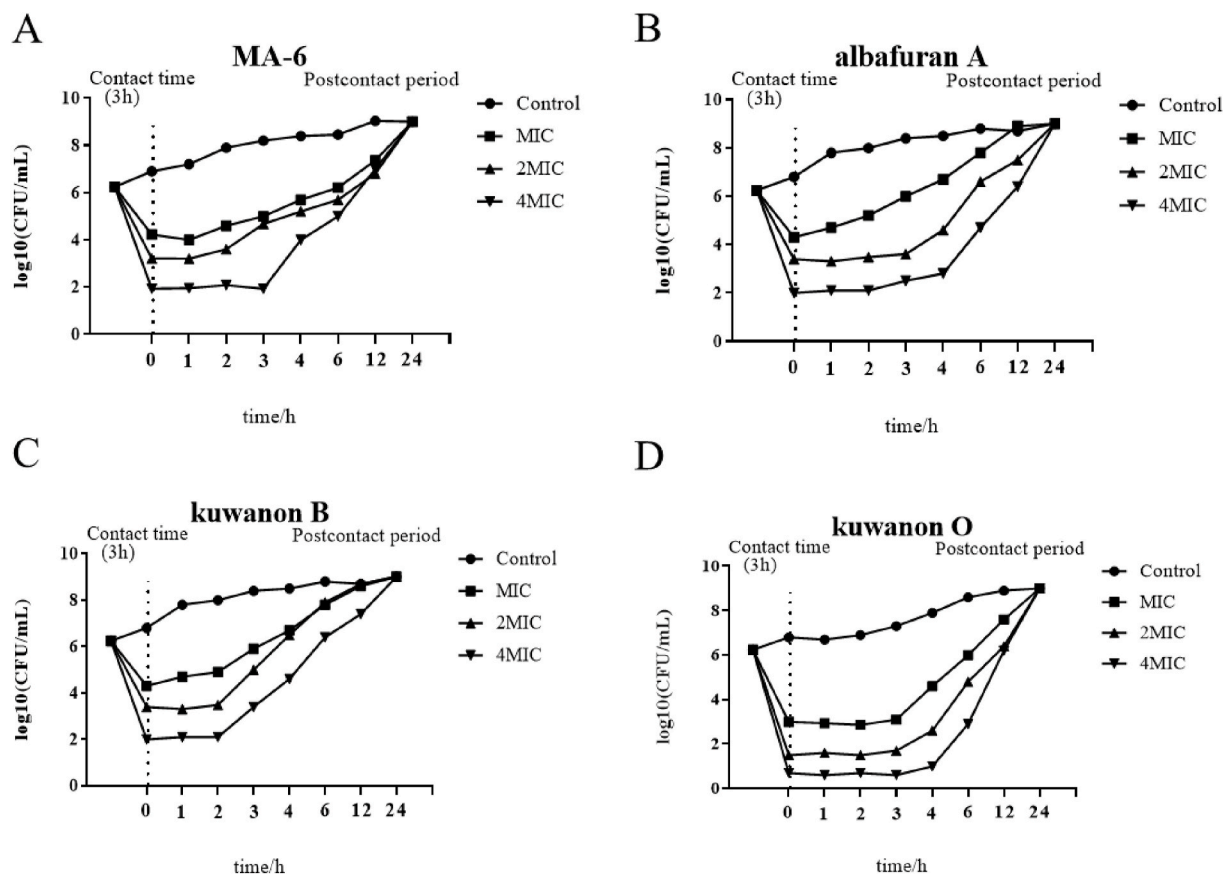


Fig. 4. Evaluation of post-contact effect (PCE). The PCE of MA-6 (A), albafrican A (B), kuwanon B (C), and kuwanon O (D) on MRSA-011 growth.

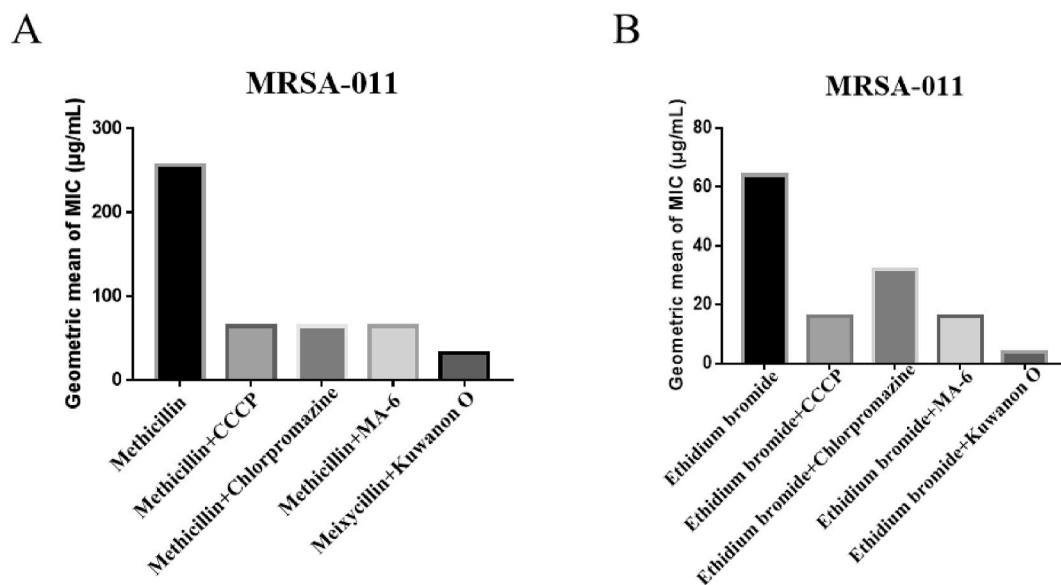


Fig. 5. MICs of the ethidium bromide (EtBr), methicillin alone, and in association with the standard inhibitor CCCP (carbonyl cyanide *m*-chlorophenylhydrazine) and CPZ (chlorpromazine), MA-6 and kuwanon O (29) against the MRSA-011.

related to the formation of bacterial resistance, the difficulty in curing chronic infectious diseases and the repeated infection of opportunistic pathogens. Fortunately, our results clearly indicated that compound kuwanon O (29) can completely destruct or remove bacterial biofilms. Therefore, it is a valuable therapy to overcome biofilm-based infections that the compound kuwanon O (29) as an adjuvant manner was

combined with conventional antibiotics.

3.4. Anti-MRSA bioactivity in vivo

To confirm whether the root bark of *M. alba* has anti-MRSA efficiency in vivo, MA-6 (40 mg/kg) were tested against MRSA in a murine acute

Table 2

MICs of the EtBr, standard inhibitors, methicillin, MA-6, kuwanon O against the MRSA-011.

Substance	MIC ($\mu\text{g/mL}$)
EtBr	64
CCCP	16
CPZ	64
methicillin	256
MA-6	64
kuwanon O	16

infection model. As shown in Fig. 7, the results were consistent with our results *in vitro*, MA-6 was obviously greater at decreasing the number of bacteria in the spleen, liver and kidney, and mortality of acute infections than the positive drug berberine chloride (Fig. 7, $p < 0.01$), indicating that five days of treatment by MA-6 led to antibacterial bioactivity *in vivo*. These results showed that the active component MA-6 of the root bark of *M. alba* at 40 mg/kg can significantly eliminate MRSA strains *in vivo*, and MA-6 can be used for treatment of deep MRSA infection.

4. Discussion

Previous studies of the root bark of *M. alba* have reported a number of pharmacological bioactivities such as anti-inflammatory (Zelová et al., 2014; Aun et al., 2013), anti-oxidative (Lan et al., 2013), anti-tumor (Shi et al., 2001; Lee et al., 2008), antiviral (Du et al., 2003), neuroprotective (Kim et al., 2011), and anti-microbial (Long et al., 2016; Wei et al., 2016). Furthermore, the root bark of *M. alba* can be used to in folk treat respiratory tract, kidney and digestive tract diseases caused by *S. aureus* (Carmine et al., 1983). In early studies, DAAs, flavonoids and stilbenes were isolated from the root bark of *M. alba*, however, studies of the anti-MRSA bioactivity of compounds were only at the tests of MICs and MBCs. Therefore, it is necessary to perform in-depth studies on the bioactivity *in vitro* and *in vivo*, and research the preliminary mechanism.

In this study, the crude extract of the root bark of *M. alba* was separated by a macro-porous resin column to obtain six fractions, here,

the MA-6 fraction was detected to have effective anti-MRSA bioactivity after susceptibility testing. Therefore, for finding more antimicrobial ingredients, further systematic phytochemical investigation was carried out according to characteristic TLC of active fraction MA-6. Chemical isolation resulted twenty-nine known compounds, most of compounds with one or more prenyl and geranyl units exhibited anti-MRSA bioactivity, these results clearly indicated that prenylation plays a crucial role in the anti-MRSA bioactivity, and prenylated compounds may be particularly effective as topic antibacterial agents. These results also confirmed that most of prenylated compounds and geranylated compounds have good hydrophobic properties, which can lead to greater penetration of cell membrane, therefore, they possess particularly effective anti-MRSA bioactivity (Chi et al., 2001). It is a great significance that compounds from medicinal plants are effective against MRSA (Jonathan et al., 2000).

Further studies on the mode of action of the compounds showed that the mode of action of compounds were different, compounds 4, 13 and 29 were bactericidal agents and they produced concentration-dependent killing, compounds 6 and 24 were antibacterial agents and they produced time-dependent killing, and these compounds had persistent antibacterial effect. The compounds with prenyl and geranyl enhances the lipophilicity, makes the compound easier to bind to the target protein through the lipid-soluble cell membrane, and greatly enhances the antibacterial activity of the compound. In addition, in this study, it was noted that natural DAA is formed through the Diels-Alder reaction, which produces excellent anti-MRSA biological activity. Preliminary research on the mechanism showed that MA-6 and kuwanon O (29) inhibited the efflux of drugs, leading to the obvious decrease in three-fold of MICs of EtBr and methicillin, and produced a same property of the EPI. moreover, compound kuwanon O (29) drastically destructed and eliminated bacterial biofilms. It was meaningful research that a single compound can act on double targets, which can inhibit the efflux of drug and the formation of biofilms, meanwhile, disperse, damage and eliminate the pre-formed biofilms, thereby effectively solved the problem of bacterial resistance. Therefore, anti-MRSA compounds in plants have irreplaceable and unique activities in destroying

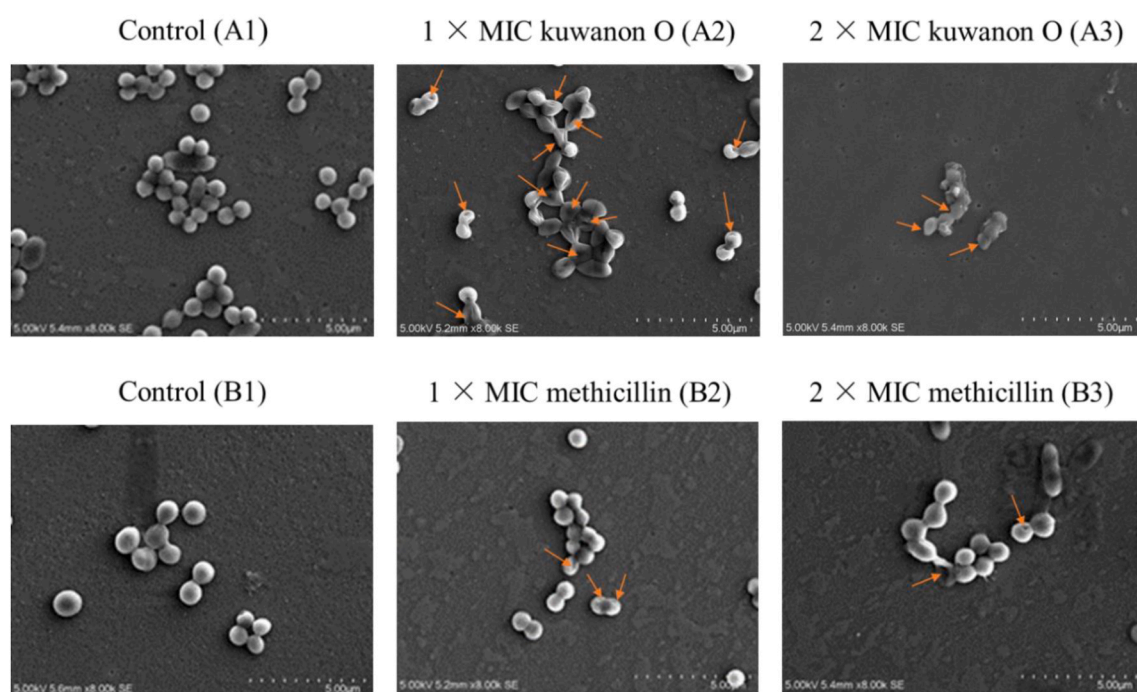


Fig. 6. SEM photography of MRSA-011. Untreated bacteria cultured for 3 h (A1, B1); bacteria treated with compound kuwanon O (29) at 1 $\times$ MIC, 2 $\times$ MIC (A2, A3); bacteria treated with positive drug methicillin at 1 $\times$ MIC, 2 $\times$ MIC (B2, B3). The red arrows represent the observed morphological changes. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

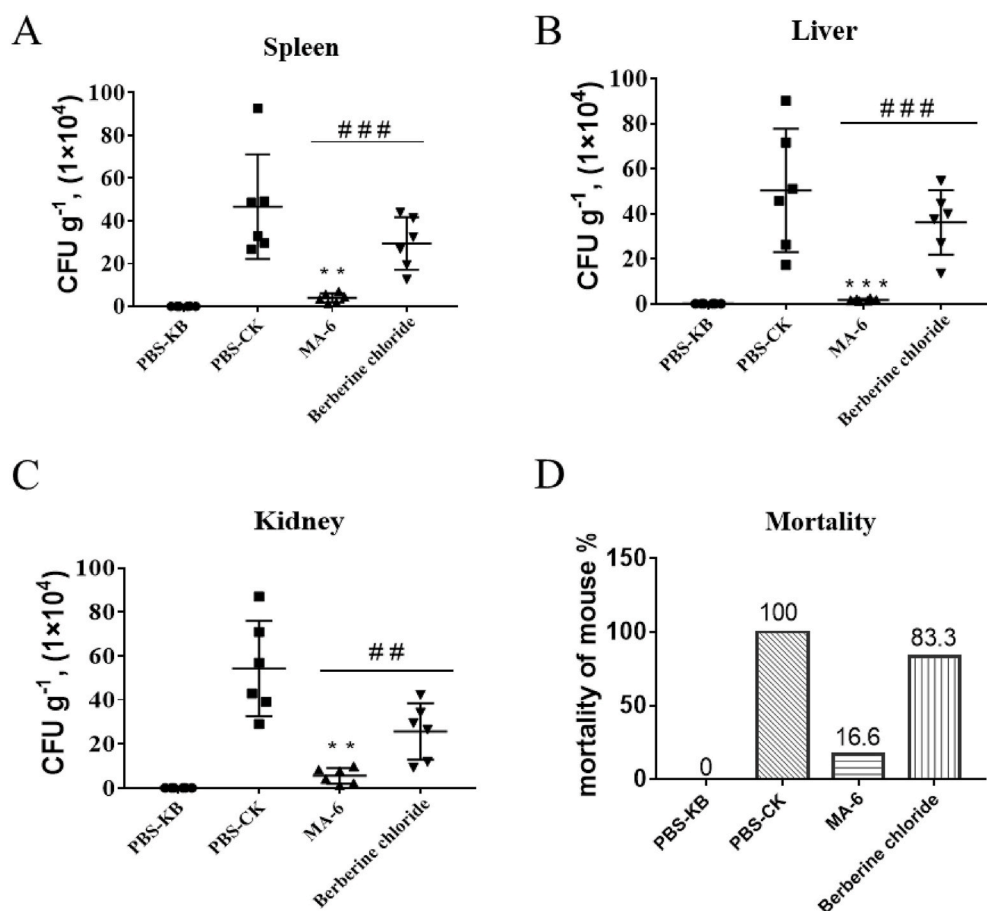


Fig. 7. The evaluation of anti-MRSA bioactivity of MA-6 in C57/BL6 mice (C57/BL6 mice peritoneal were acutely infected with MRSA-011 and fed with MA-6 (40 mg/kg), berberine chloride (50 mg/kg), and PBS by five days (once a day). Viability of the bacteria was determined after five days treatment in spleen (A), liver (B), kidney (C), and mortality (D) of acute infection. Data points represent each mouse, means SD from all mice, $n = 6$ mice per group), Values are Mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ vs CK; # $p < 0.05$, ## $p < 0.01$ and ### $p < 0.001$ vs berberine chloride.

bacterial biofilms and inhibiting the efflux of drugs.

The results of anti-MRSA tests *in vivo* showed that the bioactivity of MA-6 was obviously better at decreasing the numbers of bacteria in the spleen, liver and kidney, and mortality of acute infections than the marketed drug, berberine chloride. Meanwhile, combined with the analysis of UHPLC-QTOF MS/MS of MA-6, it was demonstrated that compounds with one or more chalcone resorcinol units, prenyl units and geranyl units indeed responsible for the anti-MRSA bioactivity of MA-6, which may explain why MA-6 was more bioactive against MRSA than other fractions. Therefore, the superior anti-MRSA bioactivity of MA-6 and compounds of the root bark of *M. alba* may be developed as a new agent to treat acute infectious diseases caused by clinical MRSA.

5. Conclusion

Our results were highly emphasized that the component MA-6 and the most compounds of the root bark of *M. alba* had better bioactivity against MRSA than antibiotics berberine, ampicillin, kanamycin both *in vitro* and *in vivo*. A series of compounds with potent anti-MRSA bioactivity obtained from the root bark of *M. alba* and the insights on mode and preliminary mechanism of action indicated that these compounds presented persistent and effective anti-MRSA bioactivity. These results may be conducive to a reduction of the consumption of antibiotics for treatment of MRSA infection, therefore, which is helping to reduce the emergence of MRSA with multi-drug resistance. In conclusion, our studies could be great meaningful for developing a new agent against MRSA and it is necessary to further study on mechanisms of action of bioactive compounds of the root bark of *M. alba*. These results will promote the safe and effective use of *M. alba* for traditional and modern medical application.

CRediT authorship contribution statement

Meng Zhu: Implemented the isolation studies, Implemented the anti-MRSA studies, Wrote the first draft, Conceptualization, Methodology, Software, Validation, Investigation. **Zhao-Jie Wang:** Implemented the anti-MRSA studies, Wrote the first draft, Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration. **Ying-Jie He:** Formal analysis. **Yan Qin:** Formal analysis. **Ying Zhou:** Implemented the anti-MRSA studies. **Zi-Heng Qi:** Formal analysis. **Zhong-Shun Zhou:** Formal analysis. **Yan-Yan Zhu:** Implemented the anti-MRSA studies. **Dan-Ni Jin:** Implemented the anti-MRSA. **Shan-Shan Chen:** Implemented the anti-MRSA studies. **Xiao-Dong Luo:** Conceived the projects, Revised manuscript and provide financial support, Revised the manuscript.

Declaration of competing interest

The authors declare no conflicts of interest regarding the publication of this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2021.114542>.

References

- Abbas, G.M., Abdel Bar, F.m., Baraka, H.N., Gohar, A.A., Lahloub, M.F., 2014. A new antioxidant stilbene and other constituents from the stem bark of *Morus nigra* L. Nat. Prod. Res. 28 (13), 952–959.
- Aelenei, P., Luca, S.V., Horhoge, C.E., Rambu, C.M., Dimitriu, G., Macovei, I., Silion, M., Aprotosoaie, A.C., Miron, A., 2019. *Morus alba* leaf extract: metabolite profiling and interactions with antibiotics against *Staphylococcus spp.* including MRSA. Phytochemistry Letters 31, 217–224.
- Archer, N.K., Mazaitis, M.J., Costerton, J.W., Leid, J.G., Powers, M.E., Shirtliff, M.E., 2011. *Staphylococcus aureus* biofilms: properties, regulation, and roles in human disease. Virulence 2 (5), 445–459.
- Aun, Z.Y., Lin, C.F., Huang, W.C., Huang, Y.L., Xu, P.Y., Chang, W.T., Wu, S.J., Liou, C.J., 2013. Anti-inflammatory effect of sophoraflavanone G isolated from *Sophora flavescens* in lipopolysaccharide-stimulated mouse macrophages. Food Chem. Toxicol. 62, 255–261.
- Balaban, N.Q., Helaine, S., Lewis, K., Ackermann, M., Aldridge, B., Andersson, D.I., Brynildsen, M.P., Bumann, D., Camilli, A., Collins, J.J., Dehio, C., Fortune, S., Ghigo, J.M., Hardt, W.D., Harms, A., Heinemann, M., Hung, D.T., Jenal, U., Levin, B. R., Michiels, J.G., Storz, M.W., Tan, T., Tenson, L., 2019. Publisher Correction: definitions and guidelines for research on antibiotic persistence. Nature Review Microbiology 17, 441–448.
- Beloin, C., Ghigo, J.M., 2008. *Escherichia coli* biofilms. Current Topic in Microbiology Immunology Ghigo 322, 249–289.
- Benincasa, M., Zahariev, S., Pelillo, C., Milan, A., Gennaro, R., Scocchi, M., 2015. PEGylation of the peptide Bac7 (1-35) reduces renal clearance while retaining antibacterial activity and bacterial cell penetration capacity. Eur. J. Med. Chem. 95, 210–219.
- Cadet, J.L., Jayanthi, S., McCoy, M.T., Ladenheim, B., Saint-Preux, F., Lehrmann, E., De, S., Becker, K.G., Brannock, C., 2013. Genome-wide profiling identifies a subset of methamphetamine (METH)-induced genes associated with METH-induced increased H4K5Ac binding in the rat striatum. BMC Genom. 14, 545.
- Cameron, D.R., Shan, Y., Zalis, E.A., Isabella, V., Lewis, K., 2018. A genetic determinant of persister cell formation in bacterial pathogens. J. Bacteriol. 200 (17), 303–318.
- Carmine, A.A., Brogden, R.N., Heel, R.C., Romankiewicz, J.A., 1983. Moxalactam (latamoxef) a review of its antibacterial activity, pharmacokinetic properties and therapeutic use. Drugs 26 (4), 279–333.
- Castro, M.A., Gamito, A.M., Tangarife-Castano, V., Zapata, B., Miguel del Corral, J.M., Mesa-Arango, A.C., Betancur-Galvis, L., San, A., 2013. Synthesis and antifungal activity of terpenyl-1,4-naphthoquinone and 1,4-anthracenedione derivatives. Eur. J. Med. Chem. 67, 19–27.
- Chan, B.M., Cbs Lau, I., Lui, S.L., Jolival, C., Ganem-Elbaz, C., Litaudon, M., Reiner, N. E., Gong, H., See, R.H., 2011. Synergistic effects of baicalein with ciprofloxacin against NorA over-expressed methicillin-resistant *Staphylococcus aureus* (MRSA) and inhibition of MRSA pyruvate kinase. J. Ethnopharmacol. 137, 767–773.
- Chi, Y.S., Jong, H.G., Son, K.H., Chang, H.W., Kang, S.S., Kim, H.P., 2001. Effects of naturally occurring prenylated flavonoids on enzymes metabolizing arachidonic acid: cyclooxygenases and lipoxygenases. Biochem. Pharmacol. 62 (9), 1185–1191.
- Conlon, B.P., Nakayasu, E.S., Fleck, L.E., Lafleur, M.D., Isabella, V.M., Coleman, K., 2013. Activated ClpP kills persisters and eradicates a chronic biofilm infection. Nature 503 (7476), 365–370.
- Dai, S.J., Chen, R.Y., Yu, D.Q., 2006a. Diels-Alder type adducts of *Morus macroura* Miq. J. China Pharm. Univ. 37, 119–122.
- Dai, S.J., De-Quan, Y.U., Zi-Ming, L., 2006b. Isolation and identification of phenolic compounds from *Morus macroura* Miq. Chin. J. Med. Chem. 16 (2), 102–105.
- Danac, R., Mangalagiu, I.L., 2014. Antimycobacterial activity of nitrogen heterocycles derivatives: bipyridine derivatives. Part III. Eur. J. Med. Chem. 74, 664–670.
- Dong-Cheol, K., Tran, Q., Hyunchoel, O., Youn-Chul, K., 2017. Steppogenin isolated from *Cudrania tricuspidata* shows antineuroinflammatory effects via NF- κ B and MAPK pathways in LPS-stimulated BV2 and primary rat microglial cells. Molecules 22 (12), 2130.
- Du, J., He, Z.D., Jiang, R.W., Ye, W.C., Xu, H.X., But, P.H., 2003. Antiviral flavonoids from the root bark of *Morus alba* L. Phytochemistry 62 (8), 1235–1238.
- Farooq, S., Wahab, A.T., Fozing, C.D.A., Rahman, A.U., Choudhary, M.I., 2015. Artonin I inhibits multidrug resistance in *Staphylococcus aureus* and potentiates the action of inactive antibiotics *in vitro*. J. Appl. Microbiol. 117 (4), 996–1011.
- Fukai, T., Hano, T., Hirakura, K., Nomura, T., Uzawa, J., Fukushima, K., 1985. Structures of two natural hypotensive Diels-Alder type adducts, mulberofurans F and G, from the cultivated mulberry tree (*Morus lhou* KOIDZ.). Chemical Pharmaceutical Bulletin 33 (8), 3195–3204.
- Fukai, T., Kaitou, K., Terada, S., 2005. Antimicrobial activity of 2-arylbenzofurans from *Morus* species against methicillin-resistant *Staphylococcus aureus*. Fitoterapia 76 (7–8), 708–711.
- Guillamet, M., Vazquez, R., Deaton, B., Shroba, J., Vazquez, L., Mercier, R.C., 2018. Host-pathogen-treatment triad host factors matter most in methicillin-resistant *staphylococcus aureus* bacteremia out-comes. Antimicrob. Agents Chemother. 62 (2), e01902–e01917.
- Gupta, D., Singh, A., Khan, A.U., 2017. Nanoparticles as efflux pump and biofilm inhibitor to rejuvenate bactericidal effect of conventional antibiotics. Nanoscale Research Letters 12, 454.
- Hano, Y., Hirakura, K., Nomura, T., Terada, S., Fukushima, K., 1984. Components of root bark of *Morus lhou* L. Structures of two new natural diels-alder adducts, kuwanons N and O. Planta Med. 50 (2), 127–130.
- He, X.M., Wu, D.L., Zou, Y.X., Wang, G.C., Ye, W.C., 2014. Study on the chemical constituents of Guangdong mulberry root bark. Modern Food Science and Technology 30 (6), 219–228.
- Heim, C.E., Bosch, M.E., Yamada, K.J., Aldrich, A.L., Kielian, T., 2020. Lactate production by *Staphylococcus aureus* biofilm inhibits HDAC11 to reprogramme the host immune response during persistent infection. Nature Microbiology 5, 1271–1284.
- Hirakata, Y., Kondo, A., Hoshino, K., Yano, H., Arai, K., Hirotani, A., Kunishima, H., Yamamoto, N., Hatta, M., Kitagawa, M., 2009. Efflux pump inhibitors reduce the invasiveness of *Pseudomonas aeruginosa*. Int. J. Antimicrob. Agents 34 (4), 343–346.
- Iinuma, M., Ohyama, M., Kawasaka, Y., Tanaka, T., 1995. Flavonoid compounds in roots of *Sophora tetraptera*. Phytochemistry 39 (3), 667–672.
- Ji, Y.K., Lee, W.S., Kim, Y.S., 2011. Isolation of cholinesterase-inhibiting flavonoids from *moras ihou*. J. Agric. Food Chem. 59, 4589–4596.
- Jiang, W.J., Ishiuchi, K., Furukawa, M., Takamiya, T., 2015. Stereospecific inhibition of nitric oxide production in macrophage cells by flavanols: synthesis and the structure-activity relationship. Bioorg. Med. Chem. 23 (21), 6922–6929.
- Jonathan, E.K., Anna, K.J., Johannes, V.S., 2000. Zulu medicinal plants with antibacterial activity. J. Ethnopharmacol. 69, 241–246.
- Jonathan, P.H., Nazia, M., Rakesh, R.G., 2012. Antimicrobial activity of PVP from an antarctic bacterium, *Janthinobacterium sp.* Ant5-2, on multi-drug and methicillin resistant *Staphylococcus aureus*. Natural Products & Bioprospecting 2 (3), 104–110.
- Jorgensen, J.H., 1991. Mechanisms of methicillin resistance in *Staphylococcus aureus* and methods for laboratory detection. Infect. Control Hosp. Epidemiol. 12 (1), 14–19.
- Kim, J.Y., Lee, W.S., Kim, Y.S., Curtis-Long, M.J., Lee, B.W., Ryu, Y.B., Park, K.H., 2011. Isolation of cholinesterase-inhibiting flavonoids from *Morus lhou*. J. Agric. Food Chem. 59 (9), 4589–4596.
- Kristian, S.A., Lauth, X., Nizet, V., Goetz, F., Neumeister, B., Peschel, A., Landmann, R., Kristian, S.A., Lauth, X., Nizet, V., 2003. Alanylation of teichoic acids protects *Staphylococcus aureus* against Toll-like receptor 2-dependent host defense in a mouse tissue cage infection model. JID (J. Infect. Dis.) 188, 414–423.
- Ku, Y.L., Liang, S.T., Lin, Y.H., Chen, L.H., Kuo, Y.Y., Ko, F.N., 2011. Anti-bacterial Use of Extract from *Morus Australis* Poir. And Compound Kuwanon H. US8071142B2.
- Lan, W.C., Tzeng, C.W., Lin, C.C., Yen, F.L., Ko, H.H., 2013. Prenylated flavonoids from *Artocarpus altilis*: antioxidant activities and inhibitory effects on melanin production. Phytochemistry 89, 78–88.
- Lee, J.C., Won, S.J., Chao, C.L., Wu, F.L., Su, C.L., 2008. Morusin induces apoptosis and suppresses NF- κ B activity in human colorectal cancer HT-29 cells. Biochem. Biophys. Res. Commun. 372 (1), 236–242.
- Letten, A.D., Hall, A.R., Levine, J.M., 2021. Using ecological coexistence theory to understand antibiotic resistance and microbial competition. Nature Ecology and Evolution 5, 431–441.
- Liu, B.R., Yan, T.N., Xiao, J., Wang10, X.L., 2018. α -Glucosidase inhibitors and antioxidants from root bark of *Morus alba*. Chinese herbal medicine 331–335.
- Long, Z., Tao, G., Jie, C., Zheng, Z.P., 2016. Characterization of a new flavone and tyrosinase inhibition constituents from the twigs of *Morus alba* L. Molecules 21 (9), 1130.
- Luo, Y., Ni, G., Guo, W., 2009. Separation and identification of medicinal chemical constituents of *Morus notabilis* branches. Sericulture Science 42 (2), 307–312.
- Mai, F., Shiota, S., Kuroda, T., Hatano, T., Tsuchiya, T., 2013. Remarkable synergies between baicalein and tetracycline, and baicalein and beta-lactams against methicillin-resistant *Staphylococcus aureus*. Microbiol. Immunol. 49 (4), 391–396.
- Mantareva, V., Veselin, K., Ivan, A., Dieter, W., 2011. Non-aggregated Ga(III)-phthalocyanines in the photodynamic inactivation of planktonic and biofilm cultures of pathogenic microorganisms. Photochem. Photobiol. Sci. 10 (1), 91–102.
- Marceau, A.H., Bernstein, D.A., Walsh, B.W., Shapiro, W., Simmons, L.A., Keck, J.L., 2013. Protein interactions in genome maintenance as novel antibacterial targets. PloS One 8 (3), e58765.
- Melander, R.J., Basak, A.K., Melander, C., 2020. Natural products as inspiration for the development of bacterial antibiofilm agents. Nat. Prod. Rep. 37 (3), 1454–1477.
- Naoya, O., Takeo, F., Jun, K., Ataru, S., Yasumasa, H., 2018. A case report of postoperative VRSA enteritis: effective management of rifampicin for vancomycin resistant *Staphylococcus aureus* enteritis after esophagectomy and colon reconstruction. International Journal of Surgery Case Reports 52, 75–78.
- Nomura, T., Hano, Y., Suzuko, S., Kohno, H., 1988. Absolute configuration of kuwanon L, a natural Diels-Alder type adduct from the *Morus* root bark. Heterocycles 27 (1), 75–82.
- Odenholt, M., Papetti, A., Gietroli, P., 2007. Antibacterial activity of red and white wine against oral streptococci. J. Agric. Food Chem. 55 (13), 5038–5042.
- Pannek, S., Higgins, P.G., Steinke, P., Jonas, D., Akova, M., Bohnert, J.A., Seifert, H., Kern, W.V., 2006. Multidrug efflux inhibition in acinetobacter baumannii: comparison between 1-(1-naphthylmethyl)-piperazine and phenyl-arginine-beta-naphthylamide. J. Antimicrob. Chemother. 57 (5), 970–974.
- Percival, S.L., Suleman, L., Vuotto, C., Donelli, G., 2015. HCAI, medical devices and biofilms: risk, tolerance and control. J. Med. Microbiol. 64 (4), 323–334.
- Qiu, Y., Hou, C., Wang, D., Sun, F., Li, X., Wang, F., Yi, H., Mu, H., Duan, J., 2018. Synthesis of hyaluronan-amikacin conjugate and its bactericidal activity against intracellular bacteria *in vitro* and *in vivo*. Carbohydr. Polym. 181, 132–140.
- Ristivojevi, P.M., Tahir, A.F., Malfent, F., Opsenica, D.M., Rollinger, M., 2019. High-performance thin-layer chromatography/bioautography and liquid chromatography-

- mass spectrometry hyphenated with chemometrics for the quality assessment of *Morus alba* samples. *Journal of Chromatography* 1594, 190–198.
- Ross, S.A., Rodriguezman, R., Radwan, M.M., Jacob, M., Ding, Y., Li, X.C., Ferreira, D., 2008. Sorocenols G and H, anti-MRSA oxygen heterocyclic Diels-Alder-type adducts from *Sorocea muriculata* roots. *J. Nat. Prod.* 71 (10), 1764–1767.
- Seong, H., Ryu, Y.B., Curtis, M.J., 2009. Tyrosinase inhibitory polyphenols from roots of *Morus lhou*. *J. Agric. Food Chem.* 57 (4), 1195–1203.
- Shen, R.C., Lin, M., 2001. Diels-Alder type adducts from *Morus cathayana*. *Phytochemistry* 57 (8), 1231–1235.
- Shi, Y.Q., Fukai, T., Sakagami, H., Chang, W.J., Yang, P.Q., Wang, F.P., Nomura, T., 2001. Cytotoxic flavonoids with isoprenoid groups from *Morus mongolica*. *J. Nat. Prod.* 64 (2), 181–188.
- Sun, D., Zhang, W., Lv, M., Yang, E., Zhao, Q., Wang, W., 2015. Antibacterial activity of ruthenium(II) polypyridyl complex manipulated by membrane permeability and cell morphology. *Bioorg. Med. Chem. Lett.* 25 (10), 2068–2073.
- Takasugi, M., Nagao, S., Masamune, T., Shirata, A., Takahashi, K., 1979. Structures of moracins E, F, G, and H, new phytoalexins from diseased mulberry. *Tetrahedron Lett.* 20 (48), 4675–4678.
- Taro, N., Yoshio, H., Masato, I., 1985. Structures of kuwanols A and B, two novel stilbene derivatives from the cultivated mulberry tree (*Morus bombycis* Koidz.). *Heterocycles* 23 (4), 819–824.
- Tintino, S.R., Cicera, D.M., Oliveira-Tintino, F.F., Raimundo, L.P., Irwin, R.A., Menezes, O.C., Tavares, J.P., Henrique, D.M., Coutinho, T.C., 2016. Evaluation of the tannic acid inhibitory effect against the NorA efflux pump of *Staphylococcus aureus*. *Microb. Pathog.* 97, 9–13.
- Tsiry, R., Quentin, L., Pierre, D., Mondher, E.J., 2015. The formation of biofilms by *Pseudomonas aeruginosa*: a review of the natural and synthetic compounds interfering with control mechanisms. *BioMed Res. Int.*, 759348.
- Ulenová, M., Sychrova, A., Hassan, S., Berchova-Bimova, K., Mejkal, K., 2019. Multiple *in vitro* biological effects of phenolic compounds from *Morus alba* root bark. *J. Ethnopharmacol.* 248, 112296.
- Vasudevan, A., Kesavan, D., Narasumani, M., Devadasan, V., Waheeta, H., Van, V., 2014. Identification of natural compound inhibitors for multidrug efflux pumps of *Escherichia coli* and *Pseudomonas aeruginosa* using in silico high-throughput virtual screening and *in vitro* validation. *PloS One* 9 (7), e101840.
- Wang, D., Lu, C., Sun, F., Cui, M., Geng, H., 2016. A tanshinone I derivative enhances the activities of antibiotics against *Staphylococcus aureus* *in vitro* and *in vivo*. *Res. Microbiol.* 168 (1), 46–54.
- Wang, D., Zhang, W., Wang, T., Li, N., Mu, H., Zhang, J., Duan, J., 2015. Unveiling the mode of action of two antibacterial tanshinone derivatives. *Int. J. Mol. Sci.* 16 (8), 17668–17681.
- Wang, Z.J., Bai, H., Lu, C.B., Hou, C., Qiu, Y.H., Zhang, P., Duan, J.Y., Mu, H.B., 2019. Light controllable chitosan micelles with ROS generation and essential oil release for the treatment of bacterial biofilm. *Carbohydr. Polym.* 205, 533–539.
- Wang, Z.J., Zhu, Y.Y., Yi, X., Zhou, Z.S., 2020. Bioguided isolation, identification and activity evaluation of antifungal compounds from *Acorus tatarinowii* Schott. *J. Ethnopharmacol.* 261, 113119.
- Wei, H., Zhu, J.J., Liu, X.Q., Feng, W.H., Wang, Z.M., Yan, L.H., 2016. Review of bioactive compounds from root barks of *Morus plants* (Sang-Bai-Pi) and their pharmacological effects. *Cogent Chemistry* 2 (1), 1212320.
- Wistrand-Yuen, P., Vrioni, G., Neou, E., Dendrinis, J., Dimitroulia, E., Poulou, A., Tsakris, A., 2011. Activity of tigecycline alone and in combination with colistin and meropenem against *Klebsiella pneumoniae* carbapenemase (KPC)-producing enterobacteriaceae strains by time-kill assay. *Int. J. Antimicrob. Agents* 37 (3), 244–247.
- Wu, H., Moser, C., Wang, H.Z., Hiby, N., Song, Z.J., 2015. Strategies for combating bacterial biofilm infections. *Int. J. Oral Sci.* 7, 1–7.
- Yang, D.S., Li, Z.L., Yang, Y.P., Xiao, W.L., Li, X.L., 2015. A new geranylated 2-aryl-benzofuran from *Morus alba*. *Chinese Herbal Medicines* 7 (2), 191–194.
- Yang, Z.G., Matsuzaki, K., Takamatsu, K., 2011. Inhibitory effects of constituents from *Morus alba* var. *multicaulis* on differentiation of 3T3-L1 cells and nitric oxide production in RAW264.7 cells. *Molecules* 16 (7), 6010–6022.
- Yang, Z., Wang, Y., Zhang, W.Y., 2012. Bioassay-guided screening and isolation of α -glucosidase and tyrosinase inhibitors from leaves of *Morus alba*. *Food Chem.* 131, 617–625.
- Zelová, H., Hanakova, Z., Cermakova, Z., Smejkal, K., Dal Acqua, S., Babula, P., Cvaka, J., Hosek, J., 2014. Evaluation of anti-inflammatory activity of prenylated substances isolated from *Morus alba* and *Morus nigra*. *J. Nat. Prod.* 77 (6), 1297–1303.
- Zheng, Z.F., Zheng, Q.J., Chen, R.Y., 2012a. Four new flavonoids from *Morus australis*. *J. Asian Nat. Prod. Res.* 14 (3), 263–269.
- Zheng, Z.P., Tan, H.Y., Wang, M., 2012b. Tyrosinase inhibition constituents from the roots of *Morus australis*. *Fitoterapia* 83 (6), 1008–1013.
- Zora, V., Marie, P., Dominik, R., Václav, S., Petr, F., Pavel, S., Jan, G., Smejkal, K., Jan, H., 2017. Prenylated flavonoid morusin protects against TNBS-induced colitis in rats. *PloS One* 12 (8), e0182464.
- Zuo, G.Y., Yang, C.X., Han, J., Li, Y.Q., Wang, G.C., 2018. Synergism of prenylflavonoids from *Morus alba* root bark against clinical MRSA isolates. *Phytomedicine* 39, 93–99.